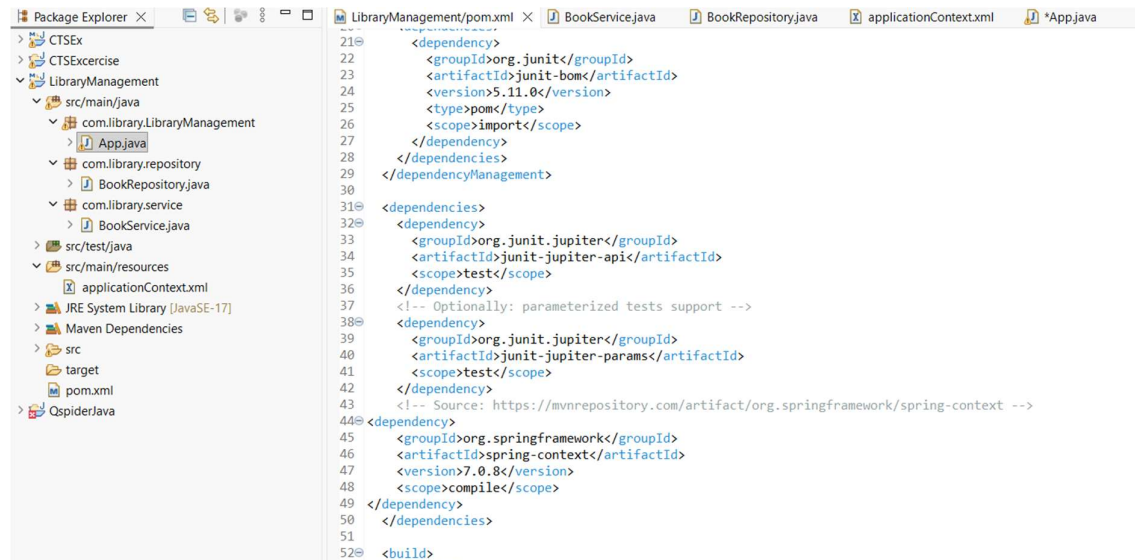


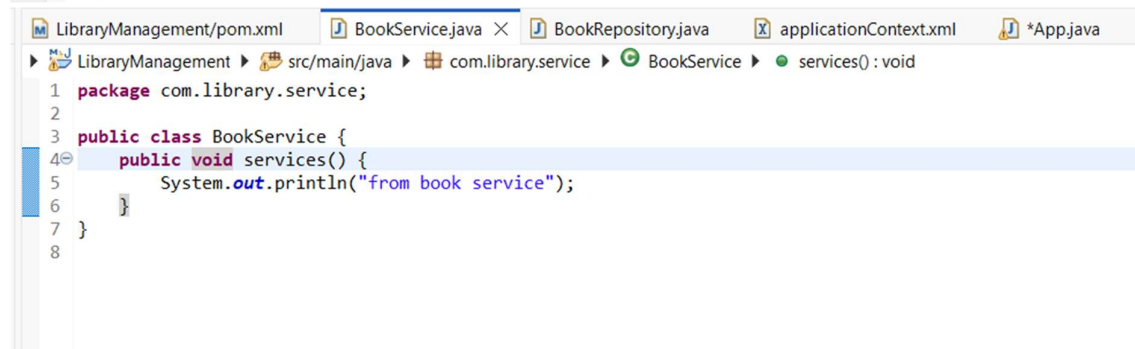
Exercise 1: Configuring a Basic Spring Application



```
21</dependency>
22<groupId>org.junit</groupId>
23<artifactId>junit-bom</artifactId>
24<version>5.11.0</version>
25<type>pom</type>
26<scope>import</scope>
27</dependency>
28</dependencies>
29</dependencyManagement>
30
31<dependencies>
32<dependency>
33<groupId>org.junit.jupiter</groupId>
34<artifactId>junit-jupiter-api</artifactId>
35<scope>test</scope>
36</dependency>
37<!-- Optionally: parameterized tests support -->
38<dependency>
39<groupId>org.junit.jupiter</groupId>
40<artifactId>junit-jupiter-params</artifactId>
41<scope>test</scope>
42</dependency>
43<!-- Source: https://mvnrepository.com/artifact/org.springframework/spring-context -->
44<dependency>
45<groupId>org.springframework</groupId>
46<artifactId>spring-context</artifactId>
47<version>7.0.8</version>
48<scope>compile</scope>
49</dependency>
50</dependencies>
51
52<build>
```



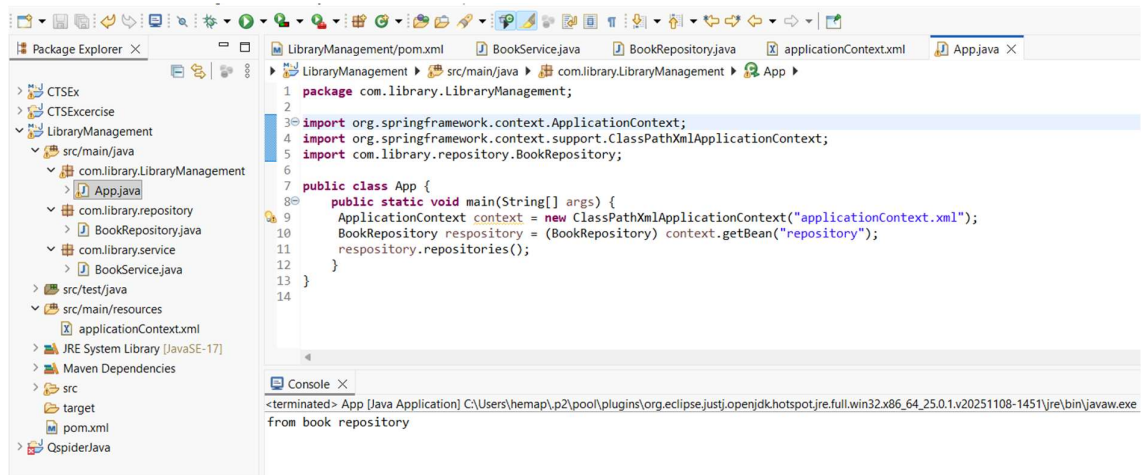
```
1 http://www.springframework.org/schema/beans/spring-beans.xsd (xsi:schemaLocation)
2<?xml version="1.0" encoding="UTF-8"?>
3<beans xmlns="http://www.springframework.org/schema/beans"
4      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5      xsi:schemaLocation="
6          http://www.springframework.org/schema/beans
7          http://www.springframework.org/schema/beans/spring-beans.xsd">
8  <bean id="service" class="com.library.service.BookService"/>
9  <bean id="repository" class="com.library.repository.BookRepository"/>
10</beans>
```



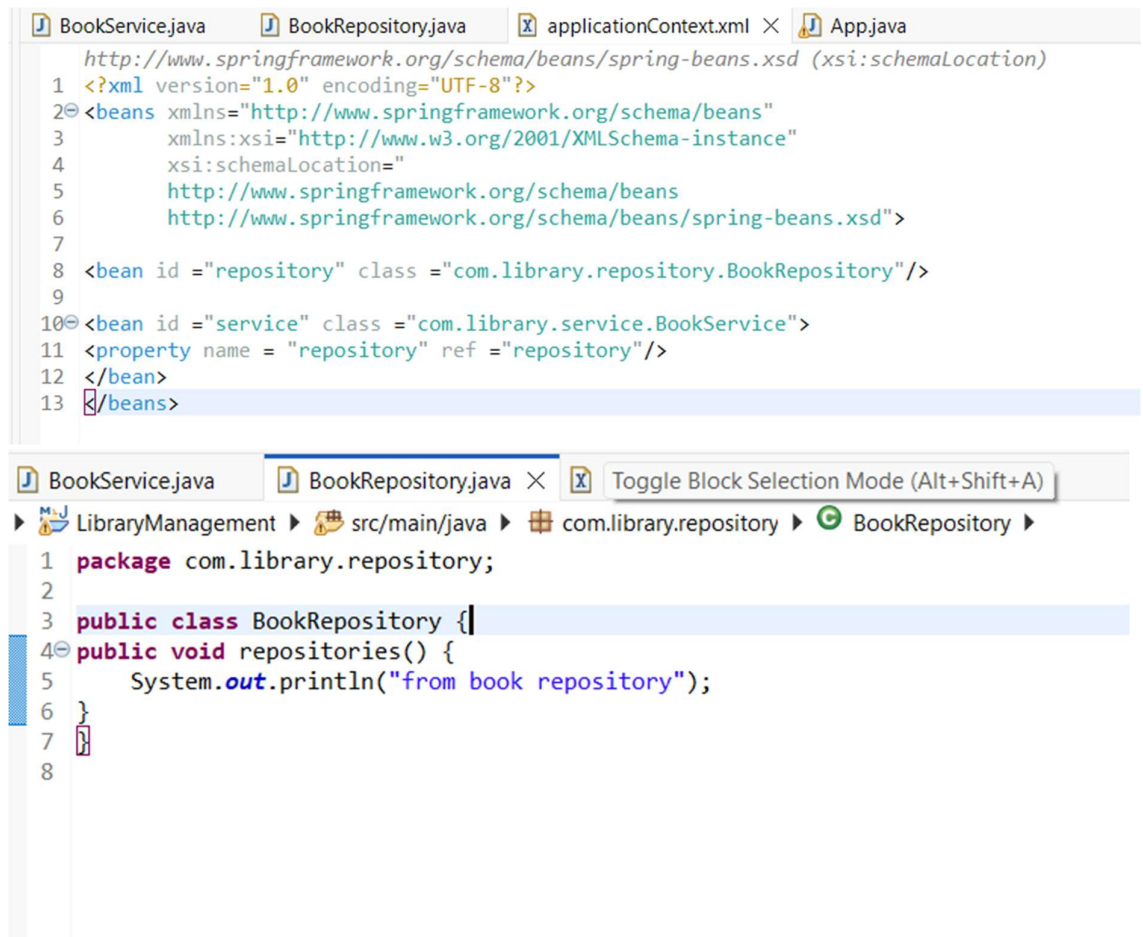
```
1 package com.library.service;
2
3 public class BookService {
4     public void services() {
5         System.out.println("from book service");
6     }
7 }
8
```



```
1 package com.library.repository;
2
3 public class BookRepository {
4     public void repositories() {
5         System.out.println("from book repository");
6     }
7 }
8
```



Exercise 2: Implementing Dependency Injection



```
BookService.java × BookRepository.java applicationContext.xml App.java ×
LibraryManagement ▶ src/main/java ▶ com.library.service ▶ BookService ▶
1 package com.library.service;
2
3 import com.library.repository.BookRepository;
4
5 public class BookService {
6     private BookRepository repository;
7
8     public void setRepository(BookRepository repository) {
9         this.repository = repository;
10    }
11
12    public void services() {
13        System.out.println("from book service");
14        repository.repositories();
15    }
16 }
17

BookService.java BookRepository.java applicationContext.xml App.java ×
LibraryManagement ▶ src/main/java ▶ com.library.LibraryManagement ▶ App ▶
1 package com.library.LibraryManagement;
2
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.support.ClassPathXmlApplicationContext;
5 import com.library.service.BookService;
6
7 public class App {
8     public static void main(String[] args) {
9         ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");
10        // BookRepository repository = (BookRepository) context.getBean("repository");
11        // repository.repositories();
12        // BookService service = (BookService) context.getBean("service");
13        BookService service = context.getBean("service", BookService.class);
14        service.services();
15    }
16 }
17 }
18

Console ×
<terminated> App [Java Application] C:\Users\hemap\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_25.0.1.v20251108-
from book service
from book repository
|
```

Exercise 4: Creating and Configuring a Maven Project

```

1 http://maven.apache.org/xsd/maven-4.0.0.xsd (xsi:schemaLocation with catalog)
2 1 <?xml version="1.0" encoding="UTF-8"?>
3 2<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4 3 xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
5 4 <modelVersion>4.0.0</modelVersion>
6 5
7 6 <groupId>com.library</groupId>
8 7 <artifactId>LibraryManagement</artifactId>
9 8 <version>0.0.1-SNAPSHOT</version>
10 9
11 10 <name>LibraryManagement</name>
12 11 <!-- FIXME change it to the project's website -->
13 12 <url>http://www.example.com</url>
14 13
15 14<properties>
16 15 <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
17 16 <!-- <maven.compiler.release>17</maven.compiler.release>-->
18 17 </properties>
19 18
20 19<dependencyManagement>
21 20<dependencies>
22 21<dependency>
23 22 <groupId>org.junit</groupId>
24 23 <artifactId>junit-bom</artifactId>
25 24 <version>5.11.0</version>
26 25 <type>pom</type>
27 26 <scope>import</scope>
28 27 </dependency>
29 28 </dependencies>
30 29 </dependencyManagement>
31 30
32 31<dependencies>
33 32<dependency>
34 33 <groupId>org.junit.jupiter</groupId>
35 34 <artifactId>junit-jupiter-api</artifactId>
36 35 <scope>test</scope>
37 36 </dependency>
38 37 <!-- Optionally: parameterized tests support -->
39 38<dependency>
40 39 <groupId>org.junit.jupiter</groupId>
41 40 <artifactId>junit-jupiter-params</artifactId>
42 41 <scope>test</scope>
43 42 </dependency>
44 43 <!-- Source: https://mvnrepository.com/artifact/org.springframework/spring-context -->
45 44<dependency>
46 45 <groupId>org.springframework</groupId>
47 46 <artifactId>spring-context</artifactId>
48 47 <version>7.0.8</version>
49 48 <scope>compile</scope>
50 49 </dependency>
51 50 <!-- Source: https://mvnrepository.com/artifact/org.springframework/spring-aop -->
52 51<dependency>
53 52 <groupId>org.springframework</groupId>
54 53 <artifactId>spring-aop</artifactId>
55 54 <version>7.0.8</version>
56 55 <scope>compile</scope>
57 56 </dependency>
58 57 <!-- Source: https://mvnrepository.com/artifact/org.springframework/spring-webmvc -->
59 58<dependency>
60 59 <groupId>org.springframework</groupId>
61 60 <artifactId>spring-webmvc</artifactId>
62 61 <version>7.0.8</version>
63 62 <scope>compile</scope>
64 63 </dependency>
65 64 </dependencies>
66 65
67 66<build>
68 67<pluginManagement><!-- lock down plugins versions to avoid using Maven defaults (may be moved to parent pom) -->
69 68<plugins>
70 69 <!-- clean lifecycle, see https://maven.apache.org/ref/current/maven-core/lifecycles.html#clean_lifecycle -->
71 70<plugin>
72 71 <artifactId>maven-clean-plugin</artifactId>
73 72 <version>3.4.0</version>
74 73 </plugin>
75 74 <!-- default lifecycle, jar packaging: see https://maven.apache.org/ref/current/maven-core/default-bindings.html#Plugin_bindings_for_jar_packaging -->

```

```

66< build>
67< pluginManagement<!-- lock down plugins versions to avoid using Maven defaults (may be moved to parent pom) -->
68< plugins>
69<!-- clean lifecycle, see https://maven.apache.org/ref/current/maven-core/lifecycles.html#clean_Lifecycle -->
70< plugin>
71< artifactId>maven-clean-plugin</artifactId>
72< version>3.4.0</version>
73</plugin>
74<!-- default lifecycle, jar packaging: see https://maven.apache.org/ref/current/maven-core/default-bindings.html#Plugin_bindings_for_jar_packaging -->
75< plugin>
76< artifactId>maven-resources-plugin</artifactId>
77< version>3.3.1</version>
78</plugin>
79< plugin>
80< artifactId>maven-compiler-plugin</artifactId>
81< version>3.13.0</version>
82< configuration>
83< source>1.8</source>
84< target>1.8</target>
85</configuration>
86</plugin>
87</plugin>
88< plugin>
89< artifactId>maven-surefire-plugin</artifactId>
90< version>3.3.0</version>
91</plugin>
92< plugin>
93< artifactId>maven-jar-plugin</artifactId>
94< version>3.4.2</version>
95</plugin>
96< plugin>
97< artifactId>maven-install-plugin</artifactId>
98< version>3.1.2</version>
99</plugin>
100< plugin>
101< artifactId>maven-deploy-plugin</artifactId>
102< version>3.1.2</version>
103</plugin>
104<!-- site lifecycle, see https://maven.apache.org/ref/current/maven-core/lifecycles.html#site_Lifecycle -->
105< plugin>
106</plugin>
</plugin>
< plugin>
< artifactId>maven-deploy-plugin</artifactId>
< version>3.1.2</version>
</plugin>
<!-- site lifecycle, see https://maven.apache.org/ref/current/maven-core/lifecycles.html#site_Lifecycle -->
< plugin>
< artifactId>maven-site-plugin</artifactId>
< version>3.12.1</version>
</plugin>
< plugin>
< artifactId>maven-project-info-reports-plugin</artifactId>
< version>3.6.1</version>
</plugin>
</plugins>
</pluginManagement>
</build>
</project>

```

Added spring AOP and spring WebMVC to the existing spring context and also inserted configuration in plugin in maven-compiler-plugin for 1.8 version compilation of the java program